

Alitronika DVS
Operational register description

Version 3.102
Date: 20 June 2006

1 OPERATIONAL REGISTER

The operational registers of Alitronika's devices are listed below. Since all register accesses are 32 bit transfers, the operational registers are grouped into four 8-bit registers.

1.1 Record(input) Configuration Registers (read/write)

Record_Config Address = H"00"

Bit	Name	Function	level
0	DVB	Selects DVB Mode	High = Enabled
1	SMP	Selects SMPTE Mode	High = Enabled
2	RAW	Select Raw Mode	High = Enabled
3	RSV	Reserved (not use in current design)	Normally Low
4	LEN	Enable loop through output	High = Enabled
7:5	GRSET	Settings for GS9060 device	Set by software
8	ETS	Enable Time Stamping	High = Enabled
9	PID	Enable PID filtering	High = Enabled
10	TEX	PID Table Exclusive	High = Exclusive Table
12:11	RSV	Reserved (not use in current design)	Normally Low
14:13	ISEL [1:0]	Select input source	"0X": USB "10": SPI "11" : ASI/Tuner
15	PCLR	Clears PID table	High = Clear PID Table
16	RSPM	SPI input mode selection	0: Constant clock mode 1: Standard DVB-SPI mode
28:17	RSV	Reserved (not use in current design)	Normally Low
29	TRST	Tuner reset	High = Reset Tuner
30	RRST	Record reset, resets input modules	High = Reset
31	REN	Record Enable	High = Enabled

1.2 Play (Output) Configuration Registers (read/write)

Play_Config Address = H"04"

Bit	Name	Function	level
0	DVB	Select DVB Mode	High = Enabled
1	SMP	Select SMPTE Mode	High = Enabled
2	SER	Select Serial output	High = Enabled
3	PAR	Select Parallel output	High = Enabled
5:4	TPS[1:0]	Select TP packet Size	"00" = 188 "01" = 204 "10" & "11" = Reserved
6	PCREST	Rest amp PCR	High = Enabled
7	ADD16	Add 16 bytes to 188 TP	High = 188+16 Mode
8	RAW	Select Raw mode	High = Enabled
9	CLK27	Select 27MHz Clock for output	High = Clk27 Selected
10	EXCLK	Select External Clock for output	High = External clock
12:11	PSPM[1:0]	SPI output mode selection	Constant Clock or Standard
13	HTP	Enable hardware generated TS	High = Enabled
14	REMUX	Enable bitrate re-multiplexing	High = Enabled
15	CTP	Select if Null/Counter packets when BRREMUX or HTP is high	Low = Null packets High= Counter packets
18:16	GPSET	Settings for GS9062 device	Set by software
19	INV	Invert the ASI output signal	High = Enabled
29:20	RSV	Reserved (not use in current design)	Normally Low
30	PRST	Play reset, resets output modules.	High = reset
31	PEN	Play Enable	High = Enabled

1.3 Output file bitrate Registers (read/write)

Output file bitrate register Address = H"08"

Bit	Name	Function	level
31:0	OFBitRate	Bitrate of the file	

1.4 Output bitrate Registers (read/write)

Output bitrate register Address = H"0C"

Bit	Name	Function	level
31:0	OBitRate	Output Bitrate	

1.5 Output burst size Registers (read/write)

Output burst size Address = H"10"

Bit	Name	Function	level
31:0	Burstsize	Output Burst size	1 to188 or 1 to 204

1.6 Status& Error Registers (read only)

Status_Error Address = H"1C"

Bit	Name	Function	level
1:0	RTPS[1:0]	Input TP Size	"00"=188, "01"=204, "10" & "11" = Reserved
2	RSYNC	Synchronised to input data stream	High = In Sync
3	RCD	Carrier Detect	High = Carrier is detected
4	RLOCK	Locked to incoming transport stream	High = Locked
5	RCNTLK	Input bitrate is estimated.	High = Bitrate estimated
6	RPCRLK	Input bitrate is obtained from PCR.	High = Bitrate from PCR
7	RINV	Input DVB-ASI signal is inverted	High = Input is inverted
8	PLOCK	Locked to output stream	High = Locked
21:15	RSV	Reserved (not use in current design)	Normally low
22	RSDFE	Record SDRAM Fifo Empty	High = Fifo Empty
23	RSDFE	Record SDRAM Fifo Full	High = Fifo Full
29:24	RSV	Reserved (not use in current design)	Normally Low
30	PSDFE	Play SDRAM Fifo Empty	High = Fifo Empty
31	PSDFE	Play SDRAM Fifo Full	High = Fifo Full

1.7.1 Bitrate Byte count Registers (read only)

Bitrate Byte count Register Address = H"20"

Bit	Name	Function	level
31:0	BrBCnt	Bitrate data counter	

1.7.2 Bitrate Time interval Registers (read only)

Bitrate time interval Register Address = H"24"

Bit	Name	Function	level
31:0	BrTime	Bitrate time interval counter	

1.8 Data Error Registers (read only)

Data Error Registers Address = H"28"

Bit	Name	Function	level
31:0	Data Err	Data error counter	

1.9 Sync Error Registers (read Only)

Sync Error Registers Address = H"2C"

Bit	Name	Function	level
31:0	Sync Err	Sync error counter	

1.10 Hardware Buffers usage indicator Registers (read only)

Record SDRAM FIFO data count register Address = H"30"

Bit	Name	Function	level
31:0	RSRAM	Number of bytes in Record SDRAM fifo	

Play SDRAM FIFO data count register Address = H"34"

Bit	Name	Function	level
31:0	PSRAM	Number of bytes in Play SDRAM fifo	

1.11.1 PID table size Registers (read only)

PID Table Size Address = H"3C"

Bit	Name	Function	level
7:0	PID size	Size of PID table	Values: 1,2,4,8,16,32,64,128
15:8	Reserved	Reserved (not use in current design)	Normally Low
23:16	PID Wdused	Position of PIDs in PID table	0- PID table size

1.11.2 PID table Registers (read/write)

PID Table Address = H"60"

Bit	Name	Function	level
12:0	PID	PID number (read/write)	
13	PID valid	PID number valid (read)	High = Valid
31-14	Reserved	Reserved (not use in current design)	Normally Low

1.12 PCI access Registers (read/write)

Play_FIFO Address = H"80"

Bit	Name	Function	level
31:0	InData	Writes into on board FIFO	

Record_FIFO Address = H"80"

Bit	Name	Function	level
31:0	OutData	Reads from on board FIFO	

1.13.1 Tuner Data Write Registers (write only)

Tuner Data Write Address = H"90"

Bit	Name	Function	level
31:0	TWData	Tuner Write Data	Used by all devices

1.13.2 Tuner Communication Control Registers (read/write)

Tuner Communication Control Address = H"94"

Bit	Name	Function	level
7:0	TunerIdx	Command register Index	
11:8	DatabyteCnt[1:0]	Number of bytes for Read/Write	Used by all devices
12	R/W	Read/Write Action Indicator	1=Read 0=Write
15:13	TunerFunction	Device Select	"0"=PLL, "1"=Satellite, "2"=Cable, "3"=Terrestrial, "4"=Annex B0, "5"=Annex B1
31:16	RSV	Reserved	Normally Low

1.13.3 Tuner Communication Status Registers (read only)

Tuner Communication Status Address = H"98"

Bit	Name	Function	level
23:0	RSV	Reserved (not use in current design)	Normally Low
24	Busy	Communication in progress	HIGH=Busy
25	Slave Nack	Error on Slave communication	HIGH=Slave Error
26	Data Nack	Error on Data communication	HIGH=Data Error
31:27	RSV	Reserved (not use in current design)	Normally Low

1.13.4 Tuner Data Read Registers (read only)

Tuner Read Data Address = H"9C"

Bit	Name	Function	level
31:0	TRData	Tuner Read Data	

2 REGISTER DESCRIPTIONS

2.1 Record Configuration Registers

The record configuration registers apply to devices that support an input (recording). The input source could be in any mode, DVB, SMPTE and RAW.

2.1.1 Record_Config Register1

RVS	RVS	RVS	LEN	RSV	RAW	SMP	DVB
-----	-----	-----	-----	-----	-----	-----	-----

Address: H"00"

Type: Write/Read

Reset: "00"

Description: This is the bits 0 to 7 of the record configuration register.

[0] DVB - Select DVB mode

Setting this bit high sets device into DVB mode.

[1] SMP - Select SMPTE mode

Setting this bit high sets device into SMPTE mode.

[2] RAW - Select RAW mode

Setting this bit high sets device into RAW mode.

[4] LEN – Enable Loop Through output

By setting this bit high loop through output is enabled.

[7:5,3] RSV - Reserved

The reserved bits are used by other Alitronika devices or are reserved for future developments and are not used in the current version.

2.1.2 Record_Config Register2

PCLR	ISEL1	ISEL0	RVS	RSV	TEX	PID	ETS
------	-------	-------	-----	-----	-----	-----	-----

Address: H"01"

Type: Write/Read

Reset: "00"

Description: This is the bits 8 to 15 of the record configuration register.

[8] ETS - Enable Time Stamping

When set high the incoming transport streams are time stamped.

The time stamp is derived from the master clock. On arrival of the 11th byte of the transport stream, the PCR byte, the content of a 32bit free running counter is taken as the time stamp.

The 32bit time stamp value is then added to the end of the transport stream.

The application must be aware of the fact that the last four bytes of the transport streams are the time stamp value not payload.

[9] PID - Enable PID filtering

Setting this bit high enables the PID filtering function.

[10] TEX – Exclusive or Inclusive PID table

In order to avoid having long PID tables, two modes of PID filtering are used.

When this bit is set high all the PIDs in the table are filtered out and when set low the PIDs in the table are kept and all other PIDs are removed.

[11:13] RSV - Reserved

The reserved bits are used by other Alitronika devices or are reserved for future developments and are not used in the current version.

[14:13] ISEL [1:0] - Input source select

These two bits form the input source select, "00" and "10" are used to indicate the device is a USB based devices and "10" selects the Parallel input and "11" the Serial input or the devices with tuners (DVB-T/S/C).

[15] PCLR – Clear PID Table

Setting this bit high clears the PID filtering table.

2.1.3 Record_Config Register3

RSV	RSV	RSV	RVS	RSV	RSV	RSV	PSPM
-----	-----	-----	-----	-----	-----	-----	------

Address: H"02"

Type: Write/Read

Reset: "00"

Description: This is the bits 16 to 23 of the record configuration register.

[16] PSPM – Selects input mode of the DVB-SPI port

The DVB standard requires that DVB-SPI should have a clock that is 1/8th of the bitrate. Often the TS source devices may use other formats. One commonly used format is a constant clock rate of 27Mhz with the "data valid" bit set high whenever there is valid data. In such cases a so-called "Target adapter" is used.

Alitronika's devices allow for selection of this format for input without the need for such an adaptor.

When this bit is set low, the standard format is selected and when set high the constant clock rate format is selected for the parallel input.

[23:17] RSV - Reserved

The reserved bits are used by other Alitronika devices or are reserved for future developments and are not used in the current version.

2.1.4 Record_Config Register4

REN	RRST	TRST	RVS	RSV	RSV	RSV	RSV
-----	------	------	-----	-----	-----	-----	-----

Address: H"03"

Type: Write/Read

Reset: "00"

Description: This is the bits 24 to 31 of the record configuration register.

[28:24] RSV - Reserved

The reserved bits are used by other Alitronika devices or are reserved for future developments and are not used in the current version.

[29] TRST – Tuner Reset

Devices with tuner on board used this bit to reset the tuner settings whenever needed.

[30] RRST – Record Section Reset

This bit is used to reset all the input modules.

[31] REN - Record Enable

When set high the device starts recording transport streams.

2.2 Play Configuration Registers

The play configuration registers apply to devices that support an output (playing).
The output could be in any mode, DVB, SMPTE and RAW.

2.2.1 Play_Config Register1

Add16	PCREST	TPS1	TPS0	PAR	SER	SMP	DVB
-------	--------	------	------	-----	-----	-----	-----

Address: H"04"

Type: Write/Read

Reset: "00"

Description: This is the 8 LSB of the Play configuration register.

The Play configuration register holds all the values needed for correct setting of the hardware sources of output section.

[0] DVB - Select DVB mode

Setting this bit high sets device into DVB mode.

[1] SMP - Select SMPTE mode

Setting this bit high sets device into SMPTE mode.

[2] SER - Select Serial output

Setting this bit high selects the Serial output.

[3] PAR - Select Parallel output

Setting this bit high selects the Parallel output.

[5:4] TPS[1:0] - Transport Packet Size

These two bits select the transport stream packet size

"00" = 188 bytes "01" = 204 bytes

"10" & "11" = Reserved

[6] PCRST - Reset PCR enabled

This bit enables the application to reset the PCR

[7] ADD16 - Add 16 bytes to a 188 byte TS

This bit enables adding of 16 bytes to 188 byte TS to make 204 byte packet size.

2.2.2 Play_Config Register2

CTP	BRREMUX	HTP	PSPM1	PSPM0	EXCLK	CLK27	RAW
-----	---------	-----	-------	-------	-------	-------	-----

Address: H"05"

Type: Write/Read

Reset: "00"

Description: This is bits 8 to 15 of the Play Configuration Register.

[8] RAW- Select Raw data mode

Setting this bit high allows the device to play back none-DVB files.

[9] CLK27 - Select 27MHz clock

When Set high the internal 27 MHz clock is selected for output data.

[10] EXCLK - Select External clock

When Set high an external is selected for output data.

[12:11] PSPM[1:0] - Play SPI mode select

"00" = Constant clock mode for 188 packet size.

"01" = Standard DVB-SPI mode for 188 packet size.

"10" = Constant clock mode for 204 (including 188 + 16 byte) packet size.

"11" = Standard DVB-SPI mode for 204 (including 188 + 16 byte) packet size.

[13] HTP - Enable Hardware Generated TS

When Set high the generation of TS by the hardware is enabled.

[14] BRREMUX - Enable Bitrate re-multiplexing

When this bit is high any output bit rate could be selected

[15] CTP - Select Null or Counter packets

When set low the hardware generates Null Packets & when set high Counter packet.

2.2.3 Play_Config Register3

RSV	RSV	RSV	RVS	INV	RSV	RSV	RSV
-----	-----	-----	-----	-----	-----	-----	-----

Address: H"06"

Type: Write/Read

Reset: "00"

Description: This is bits 16 to 23 of the Play Configuration Register.

[18:16, 23:20] RSV- Reserved

The reserved bits are used by other Alitronika devices or are reserved for future developments and are not used in the current version.

[19] INV - Invert the output ASI signal

Setting this bit high inverts the DVB-ASI output signal a useful tool for testing DVB-ASI input devices for compatibility with DVB standards which require support for both true and inverted input ASI signals.

2.2.4 Play_Config Register4

PEN	PRST	RSV	RVS	RSV	RSV	RSV	RSV
-----	------	-----	-----	-----	-----	-----	-----

Address: H"07"

Type: Write/Read

Reset: "00"

Description: This is bits 31 to 24 of the Play Configuration Register.

[29:24] RSV - Reserved

The reserved bits are used by other Alitronika devices or are reserved for future developments and are not used in the current version.

[30] PRST - Play Reset

When set high the play section modules are reset.

[31] PEN - Play Enable

When set high the device starts Playing transport streams.

2.2.5 Output File Bit Rate Register

OFileBitRate[31:24]	OFileBitRate[23:16]	OFileBitRate[15:8]	OFileBitRate[7:0]
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Address: H"08", H"09", H"0A" and H"0B"

Type: Write/Read

Reset: "00"

Description: This 32 bit register contains the output file bit rate.

The bit rate of the file played is placed in this register. When the output source is not a TS file, the content this register must be set at maximum bit rate of 216000000.

2.2.6 Output Bit Rate Register

OBRate[31:24]	OBRate[23:16]	OBRate[15:8]	OBRate[7:0]
---------------	---------------	--------------	-------------

Address: H"0C", H"0D", H"0E" and H"0F"

Type: Write/Read

Reset: "00"

Description: This 32 bit register contains the output bit rate of the transport stream.

2.2.7 Output Burst size Register

RSV [31:24]	RSV [23:16]	RSV [15:8]	OBurstSize[7:0]
-------------	-------------	------------	-----------------

Address: H"10", H"11", H"12", H"13",

Type: Write/Read

Reset: "00"

Description: This register contains the output burst size of the transport stream.

[7:0] BurstSize[7:0] – Burst Size

[31:8] RSV - Reserved

The reserved bits are used by other Alitronika devices or are reserved for future developments and are not used in the current version.

2.3 Play & Record Error & Status Registers

Applicable to all devices.

2.3.1 Record_Status Register

RINV	RPCRLK	RCNTLK	RLOCK	RCD	RSYNC	RTPS1	RTPS0
------	--------	--------	-------	-----	-------	-------	-------

Address: H"1C"

Type: Read

Reset: "00"

Description: This register contains the status of the input section.

[1:0] RTPS[1:0] - Received Transport Packet size

The packet size of the input transport stream is determined by the hardware synchronization mechanism. "00" indicated input transport packet size is 188 and when it is set "01" the input transport packet size is 204. Other conditions are currently not allowed

[2] RSYNC- Synchronised to input stream

When high it indicates the input is sync to incoming data.

[3] RCD - Carrier detect

This bit when high indicates a carrier signal is detected.

[4] RLOCK - Locked to input stream

This bit compliments bits 3 & 4 by indicating the input is now in lock & sync with the incoming data stream detected by the carrier detect.

[5] RCNTLK- Bitrate is estimated

The bit rate of the incoming Transport Stream is obtained from the PCR. In the absence of such a information (e.g. RAW data mode), the bit rate is calculated by the main controller.

[6] RPCRLK – Bitrate is obtained from PCR

This bit when set high indicated that the bit rate is obtained from the PCR.

[7] RINV- Input is Inverted DVB-ASI signal

When the input DVB-ASI signal is an inverted signal this bit is set high.

2.3.2 Play_Status Register

RSV	RSV	RSV	RSV	RSV	RSV	RSV	PLOCK
-----	-----	-----	-----	-----	-----	-----	-------

Address: H"1D"

Type: Read

Reset: 00

Description: This register contains the status of the output section.

[8] PLOCK – Play locked to the output TS

When the controller locks into the out going transport stream this bit is high to indicate that the output is in normal condition.

[15:9] RSV - Reserved

The reserved bits are either used by Alitronika's other devices or reserved for future development and are not used in the current version.

2.3.3 Record_Error Register

RSDFF	RSDFE	RSV	RSV	RSV	RSV	RSV	RSV
-------	-------	-----	-----	-----	-----	-----	-----

Address: H"1E"

Type: Read

Reset: "00"

Description: This register contains the error flags generated by the input section.

Since these bits are all error flags they are active high to indicate the error condition

[21:16] RSV - Reserved

The reserved bits are either used by Alitronika's other devices or reserved for future development and are not used in the current version.

[22] RSDFE - Record SDRAM Fifo Empty (underflow)

The SDRAM used as Fifo will set the PSDFE error flag HIGH whenever an underflow condition has occurred.

[23] RSDFF – Record SDRAM Fifo Full (overflow)

The SDRAM used as Fifo will set the PSDFF error flag HIGH whenever an overflow condition has occurred.

2.3.4 Play_Error Register

PSDFF	PSDFE	RSV	RSV	RSV	RSV	RSV	RSV
--------------	--------------	------------	------------	------------	------------	------------	------------

Address: H"1F"

Type: Read

Reset: "00"

Description: This register contains the error flags generated by the output section. Since these bits are all error flags they are active high to indicate the error condition.

[29:24] RSV - Reserved

The reserved bits are used by other Alitronika devices or are reserved for future developments and are not used in the current version.

[30] PSDFE - Play SDRAM Fifo Empty (underflow)

The SDRAM used as Fifo will set the PSDFE error flag HIGH whenever an underflow condition has occurred.

[31] PSDFF - Play SDRAM Fifo Full (overflow)

The SDRAM used as Fifo will set the PSDFF error flag HIGH whenever an overflow condition has occurred.

2.4 Record related Registers

Apply to all devices with support for input (s).

2.4.1 Bit Rate Byte Count Register

BrCnt[31:24]	BrCnt [23:16]	BrCnt [15:8]	BrCnt [7:0]
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Address: H"20", H"21", H"22" and H"23"

Type: Read

Reset: "00"

Description: This 32 bit register contains the number of data bytes received during the time period contained in the " Time Interval " register.

2.4.2 Bit Rate Time Interval Register

BrTime[31:24]	BrTime [23:16]	BrTime [15:8]	BrTime [7:0]
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Address: H"24", H"25", H"26" and H"27"

Type: Read

Reset: "00"

Description: This 32 bit register contains the time intervals during which the number of receive bytes are counted.

The bit rate of the incoming TS is normally obtained from the PCR. But there are cases that this information is not available in the PCR. In such cases the bit rate has to be estimated.

In order to be as accurate and at the same time as fast as possible to calculate the bit rate, a number of data byte is counted during a set period of time. This time interval is naturally shorter for the transport stream with higher bit rate.

The bite rate calculate using the following formula:

$$\text{BitRate} = (\text{Clk Frequency} / <\text{BrTime}>) * <\text{BRCnt}> * 8$$

2.4.3 Input Data Error Register

DataErr[31:24]	DataErr[23:16]	DataErr[15:8]	DataErr[7:0]
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Address: H"28", H"29", H"2A" and H"2B"

Type: Read

Reset: "00"

Description: This 32 bit register contains the number of data bytes errors.

Whenever the device detects an illegal code word during decoding of DVB-ASI transport stream, it generates a " Word Error " flag for the corresponding byte. A free running counter counts the number of errors. The Data Error Register contains the content of this error counter.

The Data error counter counts up for every error. There is no reset and the register may rap around. The application must compensate for these when this function is used to monitor data errors.

2.4.4 Input Sync Error Register

SyncErr[31:24]	SyncErr[23:16]	SyncErr[15:8]	SyncErr[7:0]
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Address: H"2C", H"2D", H"2E" and H"2F"

Type: Read

Reset: "00"

Description: This 32 bit register contains the number of Sync Errors.

Whenever the synchroniser, which synchronises to the incoming DVB-ASI stream can not find the sync byte, H"47", at the start of a transport packet, it increments a free running " Sync Error".

The Sync Error Register contains the content of this counter. The Sync error counter counts up for every error. There is no reset and the register may rap around. The application must compensate for these when this function is used to monitor sync errors.

2.4.5 Record SDRAM FIFO data count Register

RSDRAM[31:24]	RSDRAM [23:16]	RSDRAM [15:8]	RSDRAM [7:0]
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Address: H"30", H"31", H"32" and H"33"

Type: Read

Reset: "00"

Description: This 32 bit indicates the number of bytes in the 8 M Byte Record SDRAM.

2.5 Play related Registers

Apply to all devices with support for Output (s).

2.5.1 Play SDRAM FIFO data count Register

RSDRAM[31:24]	RSDRAM [23:16]	RSDRAM [15:8]	RSDRAM [7:0]
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Address: H"34", H"35", H"36" and H"37"

Type: Read

Reset: "00"

Description: This 32 bit indicates the number of bytes in the 8 M Byte Record SDRAM.

2.6 PID Filtering Registers

Apply to all devices with support for input (s).

2.6.1 PID Table Size Register

RSV[31:24]	PIDPos[23:16]	RSV [15:8]	PIDSize [7:0]
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Address: H"3C", H"3D", H"3E" and H"3F"

Type: Read

Reset: "00"

Description: This 32 bit indicates the number of bytes in the 8 M Byte Record SDRAM.

2.6.2 PID Table Register

RSV[31:24]	PIDPos[23:16]	RSV [15:8]	PIDSize [7:0]
------------	---------------	------------	---------------

Address: H"60", H"61", H"62" and H"63"

Type: Write/Read

Reset: "00"

Description: This 32 bit indicates the number of bytes in the 8 M Byte Record SDRAM.

The PID table is implemented by the Firmware of the main controller. The size of the PID table is represented in register H"3C". The number of PIDs available in the table can also be read at address H"3C". The PID table can be read/write sequentially. Bit 13 represents a valid PID during a read operation.

The PID filtering is controlled by 3 bits in the **Record Configuration register (H"00")**.

Bit 9 is the PID filtering enable register.

The PID table can only be accessed when this bit is low (PID filtering is disabled).

This is to stop changing the table while in use.

Bit 10 is the PID table exclusive bit. When this bit is high, all PIDs in the table are removed.

When the bit is low, the PIDs in the table are kept and all other PIDs are filtered out.

Bit 15 resets (clears) the PID table when set high.

2.7 PCI related Registers

Applicable to all PCI devices.

2.7.1 PCI Output Data Transfer Register

OutData [31:24]	OutData [23:16]	OutData [15:8]	OutData [7:0]
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Address: H"80", H"81", H"82" and H"83"

Type: Write/Read

Reset: "00"

Description: A 32 bit register used for data transfer between main controller & PCI device.

2.7.2 PCI Input Data Transfer Register

InData [31:24]	InData [23:16]	InData [15:8]	InData [7:0]
----------------	----------------	---------------	--------------

Address: H"80", H"81", H"82" and H"83"

Type: Write/Read

Reset: "00"

Description: A 32 bit register used for data transfer between main controller & PCI device.

2.8 Tuner related Registers

Apply to all devices with support for Tuner input this include all DVB-T/S/C devices.

2.8.1 Tuner Data Write Register

TWData [31:24]	TWData [23:16]	TWData [15:8]	TWData [7:0]
----------------	----------------	---------------	--------------

Address: H"90", H"91", H"92" and H"93"

Type: Write

Reset: "00"

Description: A 32 bit register used for data transfer between main controller & tuner.

This register is used by devices with tuners on board DVB-T/S/C.

2.8.2 Tuner Communication Control Register

RSV [31:16]	Type[15:13]	R/W[12]	Data [11:9]	TunerIdx[7:0]
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Address: H"94", H"95", H"96" and H"97"

Type: Write/Read

Reset: "00"

Description: A 32 bit register used for data transfer between main controller & tuner.

[7:0] Tuner Index[7:0] – Command index number

[11:8] – Number of bytes in a read/write operation

A maximum of four bytes may be written or read by each operation.

[12] R/W – Read/Write operation indicator

A High indicates a read and a Low a write operation.

[15:13] Type[2:0] – Selects the Tuner Type

The NIM on board the DVB-T/S/C device is controlled by the main controller using an I2C bus.

Each device has been allocated an address.

"0" = PLL device inside the tuner

"3" = DVB-T (Terrestrial)

"1" = DVB-S (Satellite)

"4" = DVB-C (Cable) Annex B [0]

"2" = DVB-C (Cable) Annex A

"5" = DVB-C (Cable) Annex B [1]

[31:16] RSV - Reserved

The reserved bits are used by other Alitronika devices or are reserved for future developments and are not used in the current version.

2.8.3 Tuner Communication Status Register

RSV [31:27]	Data Nack[26]	Slave Nack[25]	Busy[24]	RSV[23:0]
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Address: H"98", H"99", H"9A" and H"9B"

Type: Write/Read

Reset: "00"

Description: A 32 bit register used for data transfer between main controller & tuner.

[23:0] RSV - Reserved

The reserved bits are used by other Alitronika devices or are reserved for future developments and are not used in the current version.

[24] Busy – Communication in progress

When high it indicates that the I2C bus is busy.

[25] Slave Nack – Error on Slave Communication

When high this bit indicates there is an error in communication between the NIM (tuner) device and the main controller on the I2C bus.

[25] Data Nack – Error on Data Communication

When high there is data communication error on the I2C bus.

[31:27] RSV - Reserved

The reserved bits are used by other Alitronika devices or are reserved for future developments and are not used in the current version

2.8.4 Tuner Data Read Register

TRData [31:24]	TRData [23:16]	TRData [15:8]	TRData [7:0]
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Address: H"9C", H"9D", H"9E" and H"9F"

Type: Read

Reset: "00"

Description: A 32 bit register used for data transfer between main controller & tuner.

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